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A Fluid dynamic solver based on the Lattice Boltzmann Method

Advanced Programming for Scientific Computing Project Report

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Abstract

This project focuses on the development of a parallel fluid dynamics solver based on the Lattice Boltzmann Method (LBM), a computational approach widely used for the numerical simulation of fluid flows. Unlike traditional Computational Fluid Dynamics (CFD) methods, which solve the macroscopic conservation equations of mass, momentum, and energy, LBM describes the fluid through the evolution of particle distribution functions over a discrete lattice. The macroscopic fluid behaviour is recovered through the repeated application of collision and streaming operations, as described in the theoretical framework of the method [2, 3].

The main objective of this work is the implementation and optimization of a parallel LBM solver for the two-dimensional lid-driven cavity problem. The solver is developed using OpenMP to exploit shared-memory parallelism and improve computational performance while maintaining the accuracy of the numerical solution. The collision operator is based on a Two-Relaxation-Time (TRT) formulation, following the approach adopted in LBM implementations such as [4]. The numerical results are validated by comparing the obtained velocity field with reference benchmark data from the literature, with particular attention to the velocity profiles and the centreline values reported by Ghia et al. [1].

The project presents the theoretical foundations of the Lattice Boltzmann Method, the numerical implementation of the solver, and the parallelization strategy adopted to accelerate the computation. Furthermore, possible extensions towards distributed-memory parallelization approaches, such as MPI-based implementations, are discussed as future developments.

Keywords: Lattice Boltzmann Method, Computational Fluid Dynamics, OpenMP, Parallel Computing, Lid-Driven Cavity, Numerical Simulations

Contents

1 | LBM Model Description

The numerical model adopted in this work is based on the Lattice Boltzmann Method (LBM), using a two-dimensional D2Q9 lattice and a Two-Relaxation-Time (TRT) collision operator. Unlike traditional continuum-based numerical methods, the LBM describes the evolution of particle distribution functions defined on a discrete lattice, from which the macroscopic properties of the fluid are recovered through the moments of the distribution functions.

In the present implementation, the computational domain is discretized using a uniform Cartesian lattice. Each lattice node is associated with nine discrete velocity directions, corresponding to the D2Q9 model: one stationary direction, four orthogonal directions, and four diagonal directions. The particle distribution functions $f_i(\mathbf{x}, t)$ represent the probability density of particles moving along the discrete velocity direction $\mathbf{c}_i$.

The macroscopic density and velocity fields are obtained from the distribution functions as:

$$\rho = \sum_i f_i \tag{1.1}$$

$$\rho\mathbf{u} = \sum_i f_i \mathbf{c}_i \tag{1.2}$$

where ρ represents the fluid density and $\mathbf{u}$ is the macroscopic velocity field.

The main dimensionless parameter governing the lid-driven cavity flow is the Reynolds number:

$$Re = \frac{U_{lid} L}{\nu} \tag{1.3}$$

where U_{lid} is the velocity of the moving upper wall, L is the characteristic length of the cavity, and ν is the kinematic viscosity. In the LBM framework, the viscosity is not directly imposed, but it is controlled through the relaxation parameters of the collision operator.

In this work, the TRT collision model is employed. The distribution function is decomposed into a symmetric and an antisymmetric component, which are relaxed independently through two different parameters: ω^+ and ω^-. The kinematic viscosity is related to the relaxation parameter of the symmetric component:

$$\nu = c_s^2 \left(\frac{1}{\omega^+} - \frac{1}{2} \right) \tag{1.4}$$

where c_s is the lattice speed of sound. For the D2Q9 model:

$$c_s^2 = \frac{1}{3} \tag{1.5}$$

In the present implementation, the viscosity is computed according to the selected Reynolds number:

$$\nu = \frac{U_{lid}(N_x - 1)}{Re} \tag{1.6}$$

and the relaxation parameter ω^+ is subsequently evaluated as:

$$\omega^+ = \frac{1}{0.5 + 3\nu} \tag{1.7}$$

The second relaxation parameter, ω^-, is introduced to control the antisymmetric part of the distribution function. It is computed by imposing the TRT magic parameter condition:

$$\Lambda = \left(\frac{1}{\omega^+} - \frac{1}{2} \right) \left(\frac{1}{\omega^-} - \frac{1}{2} \right) = \frac{3}{16} \tag{1.8}$$

This additional degree of freedom is one of the main advantages of the TRT model compared with the classical single-relaxation-time BGK scheme. The independent relaxation of the symmetric and antisymmetric components allows a better control of numerical dissipation and improves the stability of simulations involving solid boundaries, such as the lid-driven cavity configuration investigated in this work.

2 | Numerical Implementation of the LBM Model

The Lattice Boltzmann model described in the previous chapter has been implemented in C++ through an object-oriented structure, separating the physical model, the computational lattice, and the boundary conditions. The main components of the implementation are represented by the classes `Lattice`, `Cavity`, and `Boundary`.

The class `Lattice` is responsible for the evolution of the particle distribution functions and for the computation of the macroscopic variables, while the class `Boundary` handles the application of the wall boundary conditions. The class `Cavity` acts as the main solver, controlling the simulation steps, the initialization procedure, the convergence monitoring, and the output of the velocity profiles.

The simulation workflow follows the classical LBM sequence:

$$\begin{aligned}&\text{Macroscopic variables} \rightarrow \text{Equilibrium distribution}\\ &\qquad\qquad\qquad\qquad \rightarrow \text{Collision} \rightarrow \text{Streaming} \rightarrow \text{Boundary treatment}\end{aligned} \tag{2.1}$$

This procedure is repeated until the velocity field reaches a steady state.

2.1. D2Q9 lattice implementation

The computational domain is discretized using a uniform Cartesian grid with $N_x \times N_y$ nodes. In the present implementation, each lattice node stores the nine distribution functions associated with the D2Q9 model.

The discrete velocity vectors are defined as:

$$\mathbf{c}_i = [(0,0),(1,0),(-1,0),(0,1),(0,-1),(1,1),(-1,-1),(-1,1),(1,-1)] \tag{2.2}$$

and the corresponding weights are:

$$w_i = \left[\frac{4}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36}\right] \tag{2.3}$$

In the C++ implementation, the distribution functions are stored in one-dimensional arrays for computational efficiency. The indexing function converts the three-dimensional information (i, x, y) into a single memory location, where i represents the lattice direction.

During the initialization procedure, the fluid is considered at rest. The equilibrium distribution is therefore computed using:

$$f_i^{eq} = \rho_0 w_i \left(1 + 3\mathbf{c}_i \cdot \mathbf{u} + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2}\mathbf{u}^2\right) \tag{2.4}$$

with:

$$\mathbf{u} = 0 \tag{2.5}$$

and $\rho_0 = 1$.

2.2. Macroscopic quantities and equilibrium computation

At each iteration, the macroscopic variables are reconstructed from the distribution functions stored in the lattice.

The density is calculated as:

$$\rho(x, y) = \sum_i f_i(x, y) \tag{2.6}$$

while the velocity components are obtained from:

$$u_x(x, y) = \frac{\sum_i f_i(x, y) c_{ix}}{\rho(x, y)} \tag{2.7}$$

$$u_y(x, y) = \frac{\sum_i f_i(x, y) c_{iy}}{\rho(x, y)} \tag{2.8}$$

These quantities are then used to compute the equilibrium distribution required by the

TRT collision step.

The equilibrium distribution implemented in the code is:

$$f_i^{eq} = \rho w_i \left[1 + 3(\mathbf{c}_i \cdot \mathbf{u}) + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2}\mathbf{u}^2 \right] \tag{2.9}$$

2.3. TRT collision implementation

The collision operator is implemented using the Two-Relaxation-Time (TRT) formulation. For each pair of opposite directions i and $\bar{i}$, the distribution function is decomposed into symmetric and antisymmetric contributions:

$$f_i^+ = \frac{1}{2}(f_i + f_{\bar{i}}) \tag{2.10}$$

$$f_i^- = \frac{1}{2}(f_i - f_{\bar{i}}) \tag{2.11}$$

The same decomposition is applied to the equilibrium distributions.

The relaxation process is then performed independently:

$$f_i^{+,new} = f_i^+ - \omega^+(f_i^+ - f_i^{eq,+}) \tag{2.12}$$

$$f_i^{-,new} = f_i^- - \omega^-(f_i^- - f_i^{eq,-}) \tag{2.13}$$

The post-collision distributions are reconstructed as:

$$f_i^{post} = f_i^{+,new} + f_i^{-,new} \tag{2.14}$$

$$f_{\bar{i}}^{post} = f_i^{+,new} - f_i^{-,new} \tag{2.15}$$

In the C++ implementation, only one direction of each opposite pair is processed during the collision step to avoid redundant calculations.

2.4. Streaming scheme

The propagation of the distribution functions is performed using a pull streaming scheme.

For each lattice direction, the distribution entering a node is obtained from the neighboring node located in the opposite direction:

$$f_i(\mathbf{x}, t+1) = f_i^{post}(\mathbf{x} - \mathbf{c}_i, t) \tag{2.16}$$

A temporary distribution array is used during the streaming procedure to avoid overwriting information that is still required during the update.

If the corresponding neighboring node is outside the computational domain, the distribution is temporarily treated through the boundary condition procedure.

2.5. Boundary conditions

The lid-driven cavity configuration is characterized by three fixed walls and one moving wall.

The bottom, left, and right walls are implemented using the bounce-back scheme. The unknown populations reaching the solid wall are reflected towards the fluid domain:

$$f_i(\mathbf{x}, t) = f_{\bar{i}}(\mathbf{x}, t) \tag{2.17}$$

The upper wall is treated as a moving boundary with velocity U_{lid}. The implementation follows the Zou-He formulation, where the unknown populations are reconstructed using the local density and the imposed wall velocity.

For the diagonal populations interacting with the moving lid, the velocity correction terms are introduced:

$$f_5 = f_6 + \frac{\rho U_{lid}}{6} \tag{2.18}$$

$$f_7 = f_8 - \frac{\rho U_{lid}}{6} \tag{2.19}$$

The four corner nodes are treated separately in order to avoid conflicts between adjacent boundary conditions.

2.6. Convergence criterion and simulation procedure

The simulation is advanced in time until a steady solution is reached. The convergence criterion is based on the relative variation of the velocity field between two consecutive iterations:

$$\epsilon = \sqrt{\frac{\sum_{x,y}[(u_x - u_x^{old})^2 + (u_y - u_y^{old})^2]}{\sum_{x,y}(u_x^2 + u_y^2) + 10^{-30}}} \tag{2.20}$$

The iterative procedure is stopped when:

$$\epsilon < 10^{-8} \tag{2.21}$$

The implementation also includes OpenMP parallelization of the most computationally intensive loops, namely the initialization, macroscopic variables calculation, equilibrium computation, collision, and streaming operations.

At the end of the simulation, the velocity profiles along the cavity centerlines are extracted and compared with the reference data reported by Ghia et al. for the validation of the numerical model.

2.6.1. LBM simulation algorithm

The complete numerical procedure adopted for the lid-driven cavity simulation can be summarized through the following algorithm. The algorithm describes the sequence of operations performed at each time step of the Lattice Boltzmann solver, from the initialization of the distribution functions to the convergence of the velocity field.

Algorithm 2.1 D2Q9-TRT LBM algorithm for the lid-driven cavity problem

Define the computational domain and discretize the cavity using a uniform lattice

Set the physical parameters: Re, U_{lid}, cavity length L, and reference density ρ_0

Compute the kinematic viscosity:

$$\nu = \frac{U_{lid} L}{Re}$$

Compute the TRT relaxation parameters: ω^+ and ω^-

Initialize the density and velocity fields

Initialize all distribution functions with the equilibrium distribution

while the convergence criterion is not satisfied **do**

  Compute the macroscopic density and velocity fields from the distribution functions

  Evaluate the equilibrium distribution functions

  Apply the TRT collision operator:

  - decompose the distributions into symmetric and antisymmetric components
  - relax the two components using ω^+ and ω^-
  - reconstruct the post-collision distributions

  Perform the streaming step by propagating the post-collision distributions along the discrete lattice directions

  Apply the boundary conditions:

  - bounce-back condition on the stationary walls
  - moving-wall condition on the upper lid

  Update the velocity field

  Evaluate the relative variation of the velocity field between two consecutive iterations

end while

Extract the velocity profiles along the cavity centerlines

Compare the numerical results with reference benchmark data

3 | Numerical Results

This chapter presents the numerical results obtained with the developed Lattice Boltzmann solver for the two-dimensional lid-driven cavity problem. The numerical solutions are compared against the benchmark data published by Ghia et al. [1], which represent one of the most widely accepted reference datasets for this benchmark configuration.

All simulations were performed on the same computational grid composed of 129×129 lattice nodes. The lid velocity was kept constant throughout all simulations and equal to

$$U_{lid} = 0.05,$$

while different Reynolds numbers were obtained by varying the kinematic viscosity according to

$$Re = \frac{U_{lid} L}{\nu},$$

where L denotes the characteristic cavity length expressed in lattice units.

Maintaining the same spatial discretization and the same lid velocity for every simulation allows the influence of the Reynolds number to be investigated without introducing additional numerical variables. The comparison therefore highlights only the effect of the changing flow regime.

Before comparing the numerical results with the reference benchmark, the velocity components were non-dimensionalized according to

$$u^* = \frac{u}{U_{lid}}, \qquad v^* = \frac{v}{U_{lid}},$$

thus obtaining the same quantities reported by Ghia et al. [1]. This normalization makes the comparison independent of the particular value of the lid velocity employed in the present simulations.

For each Reynolds number, the comparison is carried out using the two standard centreline velocity profiles adopted in the original benchmark:

- the normalized horizontal velocity u/U_{lid} evaluated along the vertical centreline of the cavity;
- the normalized vertical velocity v/U_{lid} evaluated along the horizontal centreline.

Throughout this chapter, the benchmark data from Ghia et al. [1] are represented by the **orange curves**, while the numerical results obtained with the present implementation are represented by the **blue curves**.

3.1. Comparison with the benchmark solution

The comparison is presented for the following Reynolds numbers:

$$Re = 100,\ 400,\ 1000,\ 3200,\ 5000,\ 7500.$$

For all these operating conditions, the adopted numerical formulation provides stable and physically meaningful solutions.

The agreement between the present numerical results and the benchmark solution is excellent for the lowest Reynolds numbers. Both the horizontal and vertical velocity profiles closely follow the reference curves over the entire computational domain, with only negligible differences.

As the Reynolds number increases, progressively larger discrepancies become visible, especially in those portions of the profiles where the velocity gradients are more pronounced. Nevertheless, the overall shape of both normalized velocity components remains in good agreement with the benchmark data up to $Re = 7500$, confirming the correctness of the implemented collision operator, streaming procedure and boundary treatment.

The benchmark dataset also includes the case

$$Re = 10000,$$

which represents the highest Reynolds number considered by Ghia et al. [1]. However, for the numerical configuration adopted in this work (D2Q9 lattice, TRT collision operator, relaxation parameters derived from the prescribed Reynolds number and a 129×129 computational grid), the iterative procedure does not converge towards a physically meaningful steady solution. Consequently, the computed macroscopic variables lose their physical significance, making any quantitative comparison with the benchmark inappropriate.

For this reason, the numerical solution corresponding to $Re = 10000$ is intentionally omitted from the comparison. Instead, only the reference profile published by Ghia et al. [1] is reported in order to complete the sequence of benchmark cases and to provide the expected velocity distributions at this Reynolds number.

The numerical reasons underlying this limitation are discussed in detail in the following chapter, where the influence of the relaxation parameters, lattice resolution and boundary treatment on the stability of the adopted Lattice Boltzmann formulation is analysed.

3.2. Velocity profile comparisons

Figures 3.1–3.7 summarize the comparison between the numerical results and the benchmark solution.

Each figure corresponds to one Reynolds number and contains two plots:

- the normalized horizontal velocity component u/U_{lid} along the vertical centreline;
- the normalized vertical velocity component v/U_{lid} along the horizontal centreline.

For the Reynolds numbers ranging from 100 to 7500, each figure reports both the benchmark data (orange curves) and the numerical results obtained with the present TRT-LBM implementation (blue curves).

The final figure corresponds to the benchmark case $Re = 10000$. Since the present implementation does not provide a stable and physically acceptable solution for this operating condition, only the reference profiles published by Ghia et al. [1] are shown. This choice provides a complete overview of the benchmark dataset while avoiding the presentation of non-converged numerical results.

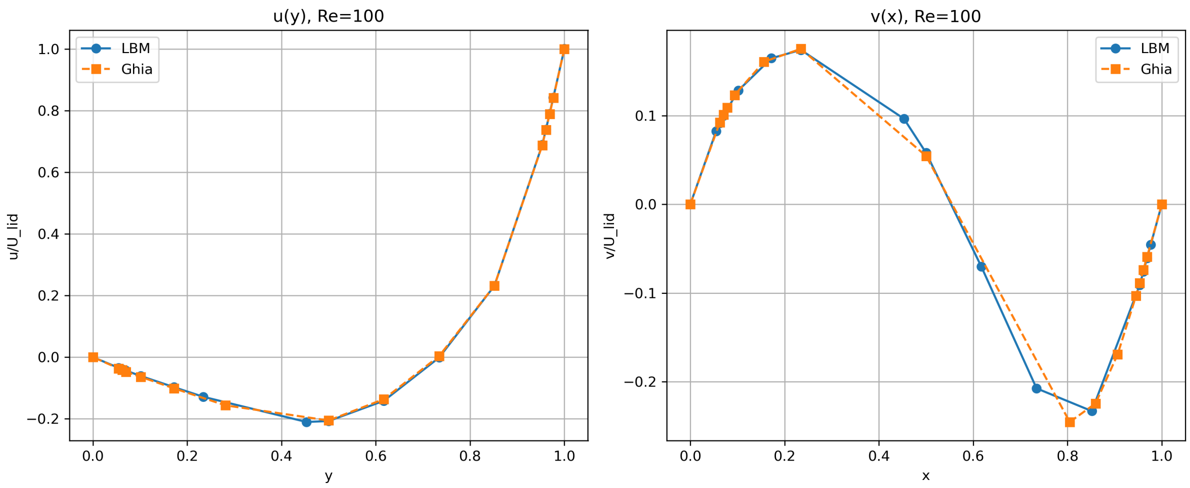

Figure 3.1: Comparison between the normalized velocity profiles obtained with the present TRT-LBM solver (blue) and the benchmark data of Ghia et al. [1] (orange) for $Re = 100$. The left panel reports the horizontal velocity profile $U(y) = u/U_{lid}$ evaluated along the vertical centreline of the cavity, while the right panel reports the vertical velocity profile $V(x) = v/U_{lid}$ evaluated along the horizontal centreline.

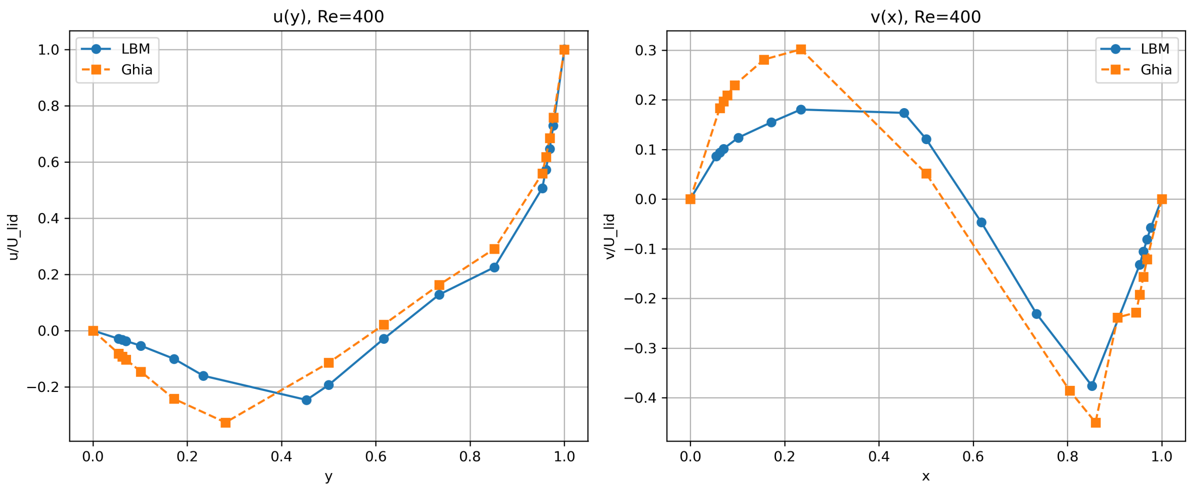

Figure 3.2: Comparison between the normalized velocity profiles obtained with the present TRT-LBM solver (blue) and the benchmark data of Ghia et al. [1] (orange) for $Re = 400$. The left panel reports the horizontal velocity profile $U(y) = u/U_{lid}$ along the vertical centreline, whereas the right panel reports the vertical velocity profile $V(x) = v/U_{lid}$ along the horizontal centreline.

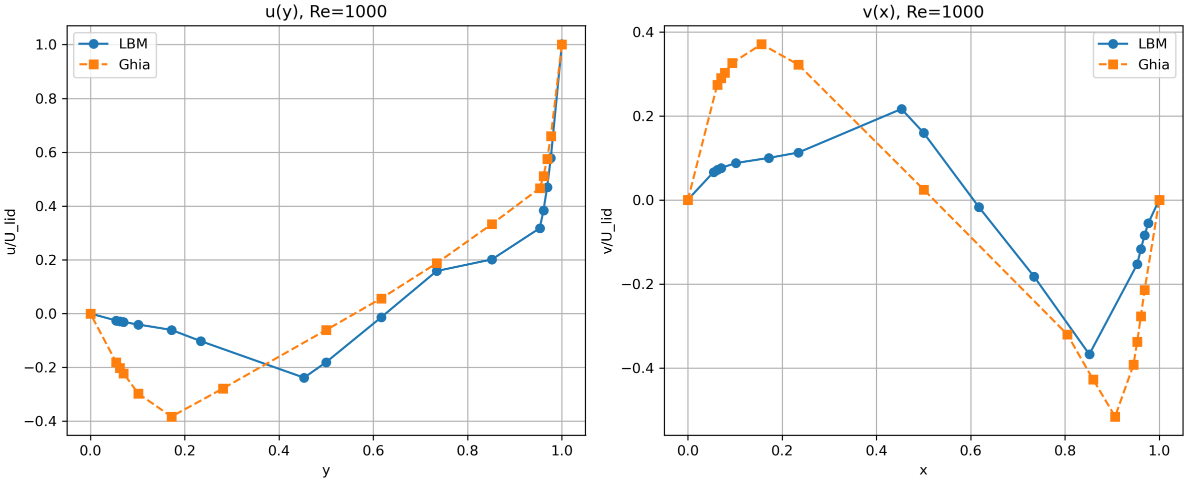

Figure 3.3: Comparison between the normalized velocity profiles obtained with the present TRT-LBM solver (blue) and the benchmark data of Ghia et al. [1] (orange) for $Re = 1000$. The left panel reports the horizontal velocity profile $U(y) = u/U_{lid}$ along the vertical centreline, while the right panel reports the vertical velocity profile $V(x) = v/U_{lid}$ along the horizontal centreline.

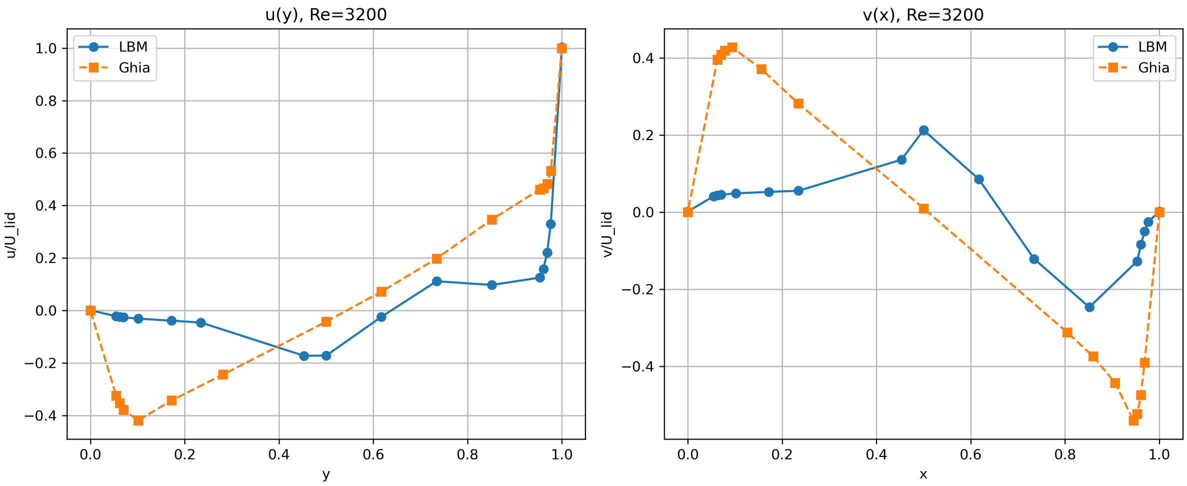

Figure 3.4: Comparison between the normalized velocity profiles obtained with the present TRT-LBM solver (blue) and the benchmark data of Ghia et al. [1] (orange) for $Re = 3200$. The left panel reports the horizontal velocity profile $U(y) = u/U_{lid}$ evaluated along the vertical centreline, whereas the right panel reports the vertical velocity profile $V(x) = v/U_{lid}$ evaluated along the horizontal centreline.

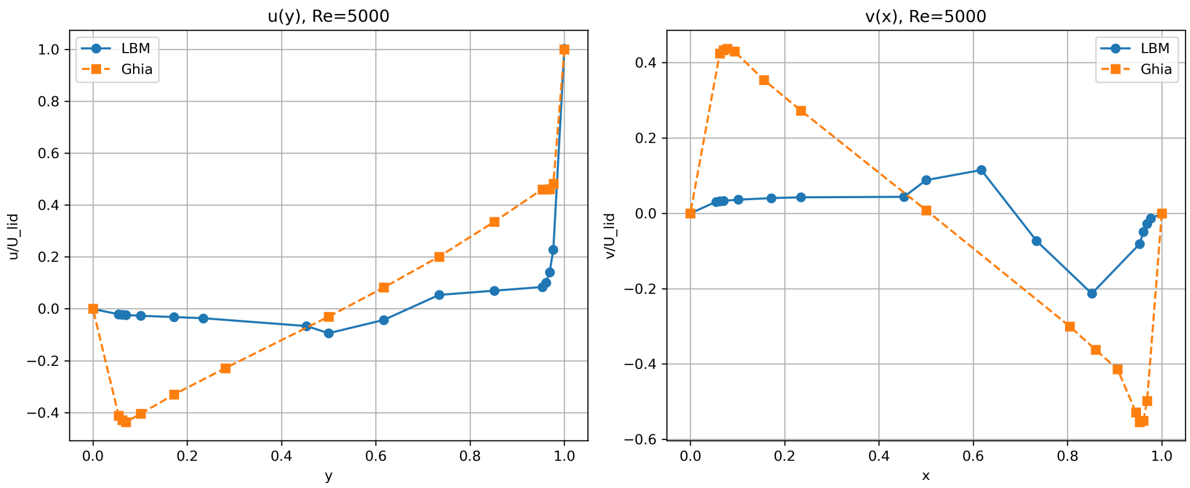

Figure 3.5: Comparison between the normalized velocity profiles obtained with the present TRT-LBM solver (blue) and the benchmark data of Ghia et al. [1] (orange) for $Re = 5000$. The left panel reports the horizontal velocity profile $U(y) = u/U_{lid}$ evaluated along the vertical centreline, while the right panel reports the vertical velocity profile $V(x) = v/U_{lid}$ evaluated along the horizontal centreline.

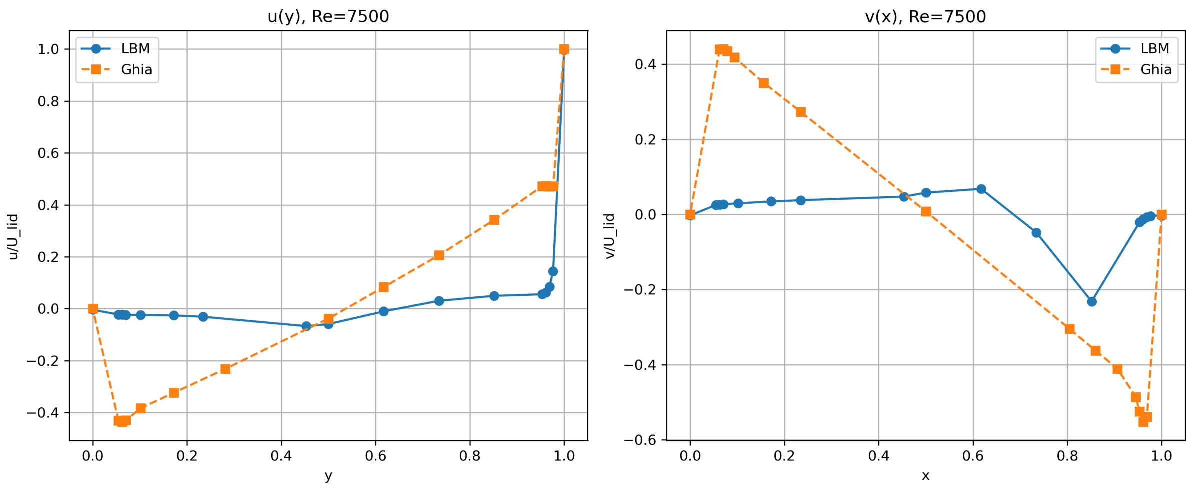

Figure 3.6: Comparison between the normalized velocity profiles obtained with the present TRT-LBM solver (blue) and the benchmark data of Ghia et al. [1] (orange) for $Re = 7500$. The left panel reports the horizontal velocity profile $U(y) = u/U_{lid}$ evaluated along the vertical centreline, whereas the right panel reports the vertical velocity profile $V(x) = v/U_{lid}$ evaluated along the horizontal centreline.

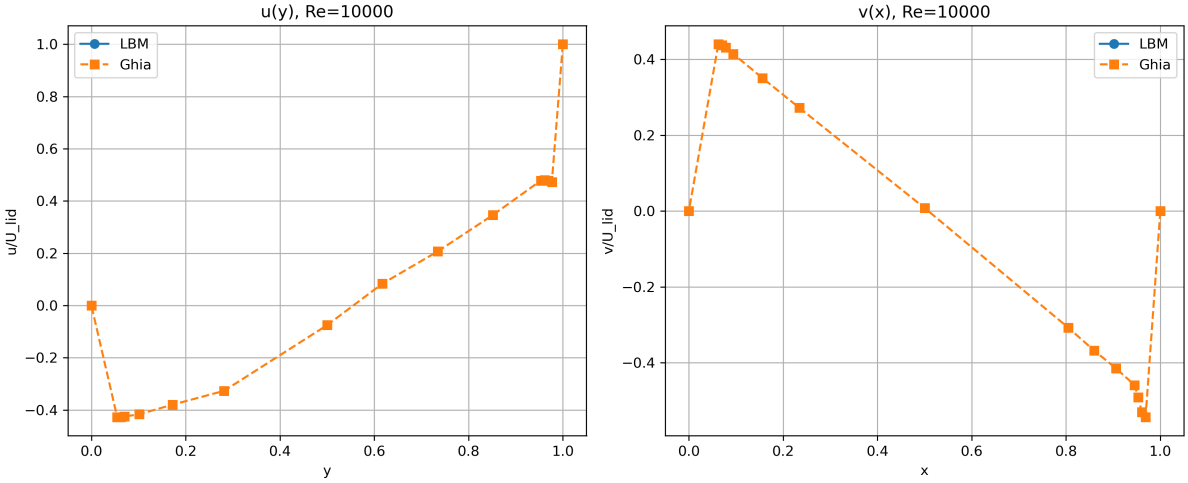

Figure 3.7: Reference benchmark velocity profiles for $Re = 10000$ reported by Ghia et al. [1]. The left panel reports the normalized horizontal velocity profile $U(y) = u/U_{lid}$ along the vertical centreline, whereas the right panel reports the normalized vertical velocity profile $V(x) = v/U_{lid}$ along the horizontal centreline. The numerical solution obtained with the present implementation is intentionally omitted because the adopted TRT-LBM configuration does not converge to a physically meaningful steady-state solution under these operating conditions.

4 | Validity limits of the adopted LBM configuration at high Reynolds numbers

The analysis of the simulations performed at high Reynolds numbers requires a careful evaluation of the relationship between the physical flow parameters and the numerical parameters of the adopted Lattice Boltzmann formulation.

The results obtained in this work show that increasing the Reynolds number does not only modify the physical characteristics of the flow, but also progressively challenges the assumptions and stability conditions of the numerical scheme. Therefore, the loss of stability observed for the highest Reynolds number cases must be interpreted as a limitation of the specific numerical configuration adopted in this work, rather than as an intrinsic limitation of the Lattice Boltzmann Method itself.

The main factors affecting the validity range of the simulations are the relaxation parameters of the collision operator, the spatial resolution of the lattice, and the accuracy of the boundary treatment in regions characterized by strong velocity gradients.

4.1. Effect of Reynolds number on the relaxation parameter

The Reynolds number governing the lid-driven cavity flow is defined as:

$$Re = \frac{U_{lid} L}{\nu} \tag{4.1}$$

where U_{lid} is the velocity imposed at the moving wall, L is the characteristic length of the cavity, and ν is the kinematic viscosity.

In the present implementation, the viscosity is directly related to the relaxation parameter of the TRT collision model. In lattice units, the kinematic viscosity is expressed as:

$$\nu = c_s^2 \left(\frac{1}{\omega_+} - \frac{1}{2} \right) \tag{4.2}$$

where c_s is the lattice speed of sound and ω_+ is the relaxation frequency associated with the symmetric component of the distribution functions.

Consequently, increasing the Reynolds number while maintaining a fixed lattice size and lid velocity requires a reduction of the kinematic viscosity. This leads to a progressive increase of the relaxation frequency:

$$Re \uparrow \quad \Longrightarrow \quad \nu \downarrow \quad \Longrightarrow \quad \omega_+ \to 2 \tag{4.3}$$

The value $\omega_+ = 2$ represents the limit condition corresponding to the minimum relaxation time. Although the LBM equations remain formally defined in this regime, values of ω_+ close to this limit represent a critical operating condition.

Indeed, when the relaxation parameter approaches two, the collision operator becomes strongly over-relaxed. The numerical dissipation introduced by the collision process is progressively reduced and small errors generated by the discretization, floating-point arithmetic, or boundary treatment may be amplified rather than dissipated.

Therefore, high Reynolds number simulations require a compromise between the physical requirement of low viscosity and the numerical requirement of maintaining relaxation parameters sufficiently far from their stability limits.

4.2. Influence of spatial resolution and boundary layer representation

The spatial resolution of the computational lattice is a fundamental factor in determining the accuracy of high Reynolds number simulations.

For a characteristic lattice resolution of N nodes, the lattice spacing is:

$$\Delta x = \frac{L}{N-1} \tag{4.4}$$

The velocity gradients near solid walls are associated with the development of boundary layers. A characteristic estimation of the boundary layer thickness is:

$$\delta \sim \frac{L}{\sqrt{Re}} \tag{4.5}$$

This relation shows that increasing the Reynolds number produces thinner boundary layers because viscous diffusion becomes less effective in spreading momentum away from the wall.

The number of lattice points contained inside the boundary layer can be estimated as:

$$N_\delta = \frac{\delta}{\Delta x} \tag{4.6}$$

For moderate Reynolds numbers, the value of N_δ remains large enough to properly describe the velocity variation close to the wall. In this condition, the lattice is able to reproduce the main physical features of the cavity flow.

Conversely, when the Reynolds number becomes sufficiently large, the boundary layer thickness approaches the order of magnitude of the lattice spacing. As a consequence, only a few lattice nodes are available to represent regions where the velocity changes rapidly.

This under-resolution produces several effects:

- inaccurate representation of wall shear stresses;
- incorrect prediction of vorticity generation near the walls;
- increased sensitivity to the collision parameters;
- artificial numerical dissipation or oscillations.

Therefore, increasing the Reynolds number without increasing the lattice resolution does not provide a more accurate representation of the same physical problem. Instead, it changes the numerical regime of the simulation by forcing the method to approximate flow structures smaller than the available grid scale.

4.3. Boundary effects and near-wall accuracy

The treatment of solid boundaries represents another critical aspect of Lattice Boltzmann simulations.

In the present implementation, the stationary walls are treated using a bounce-back scheme, while the moving lid condition is introduced by correcting the incoming particle distributions according to the imposed wall velocity.

The bounce-back method approximates the no-slip condition by reversing the momentum of particle populations interacting with the solid wall. This approach is accurate when the velocity field varies smoothly over the distance of a lattice cell.

At high Reynolds numbers, however, the velocity gradients close to the wall become increasingly large. The physical boundary layer may become comparable with the distance between adjacent lattice nodes, making the discrete representation of the wall less accurate.

The effect is particularly relevant near the cavity corners, where the interaction between two solid boundaries generates strong local gradients and secondary vortical structures. These regions are among the most demanding parts of the flow field because small discretization errors may significantly modify the local vorticity distribution.

Therefore, for high Reynolds number flows, accurate wall modelling and sufficient grid refinement become essential to correctly reproduce the physical momentum transfer between the fluid and the boundaries.

4.4. Mach number constraint and weak compressibility effects

Another important aspect of LBM simulations is related to the weakly compressible formulation adopted by the method.

The Lattice Boltzmann approach recovers the incompressible Navier–Stokes equations under the assumption of low Mach number:

$$Ma = \frac{U}{c_s} \ll 1 \tag{4.7}$$

where U is the characteristic flow velocity and c_s is the lattice speed of sound.

Maintaining a sufficiently small Mach number ensures that density variations remain negligible and that compressibility effects do not influence the solution.

However, increasing the lid velocity to reach larger Reynolds numbers may violate this assumption. Therefore, increasing Re by modifying velocity instead of viscosity can introduce additional compressibility errors.

For this reason, high Reynolds number simulations generally require a careful selection of both velocity scale and viscosity, together with adequate grid resolution.

4.5. Discussion of the present numerical configuration

The simulations performed in this work use a 129×129 lattice and a TRT collision model. This configuration provides reliable results for low and moderate Reynolds numbers, where the relaxation parameters remain within a stable range and the main flow structures are adequately resolved.

In this regime, the comparison with reference solutions such as the benchmark data reported by Ghia et al. is physically meaningful because the numerical assumptions of the adopted LBM model are satisfied.

For very high Reynolds numbers, the simultaneous occurrence of several effects limits the reliability of the obtained solution:

1. the viscosity required by the target Reynolds number forces the relaxation parameter ω_+ close to its upper stability limit;
2. the fixed lattice resolution becomes insufficient to resolve thin boundary layers and strong near-wall gradients;
3. the discrete boundary treatment becomes increasingly sensitive to unresolved flow structures.

Therefore, the numerical instability observed in the highest Reynolds number simulations should not be interpreted as a failure of the Lattice Boltzmann Method, but rather as the consequence of applying a specific LBM configuration outside its practical range of validity.

Extending the simulation capability towards higher Reynolds numbers would require additional numerical improvements, such as lattice refinement, improved boundary schemes, optimized relaxation parameters, or more advanced collision models specifically designed for high-Reynolds-number applications.

5 | Software implementation

The numerical solver developed in this work has been implemented in C++ following an object-oriented structure. The aim of this organization is to separate the physical model, the lattice operations, the boundary conditions, and the high-level simulation control.

The implementation is divided into three main classes: `Lattice`, `Boundary`, and `Cavity`. The `Lattice` class contains the numerical representation of the D2Q9 lattice and manages the evolution of the particle distribution functions. The `Boundary` class is responsible for the application of the wall boundary conditions, while the `Cavity` class coordinates the complete simulation procedure.

This modular structure allows each physical process of the LBM method to be independently managed and facilitates future extensions of the solver.

5.1. Lattice class implementation

The `Lattice` class represents the core component of the LBM solver. It contains the discrete velocity model, the lattice weights, the distribution functions, and all the operations required for the evolution of the particle populations.

The class stores three distribution fields:

- f_i, representing the current particle distribution;
- f_i^{eq}, representing the local equilibrium distribution;
- f_i^{post}, representing the distribution after collision and before streaming.

In addition, arrays containing the macroscopic variables are stored:

- density ρ;
- velocity components u_x and u_y.

The lattice dimensions and TRT relaxation parameters ω^+ and ω^- are also stored as class parameters.

5.1.1. Lattice initialization

The initialization procedure assigns the initial macroscopic state of the fluid and computes the corresponding equilibrium distribution.

For the lid-driven cavity simulations, the initial condition is a fluid at rest:

$$u_x = u_y = 0 \tag{5.1}$$

with uniform density:

$$\rho = \rho_0 \tag{5.2}$$

The distribution functions are initialized according to the equilibrium function:

$$f_i = f_i^{eq} \tag{5.3}$$

This provides a physically consistent initial state before starting the time evolution.

5.1.2. Macroscopic variables computation

At each iteration, the macroscopic variables are reconstructed from the distribution functions.

The density is computed as:

$$\rho = \sum_i f_i \tag{5.4}$$

while the velocity components are obtained through:

$$u_x = \frac{\sum_i f_i c_{ix}}{\rho} \tag{5.5}$$

$$u_y = \frac{\sum_i f_i c_{iy}}{\rho} \tag{5.6}$$

These quantities represent the link between the microscopic particle description and the macroscopic fluid behaviour.

5.1.3. Equilibrium distribution computation

The equilibrium distribution is evaluated using the second-order expansion of the Maxwell–Boltzmann distribution:

$$f_i^{eq} = \rho w_i \left[1 + 3\mathbf{c}_i \cdot \mathbf{u} + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2}\mathbf{u}^2 \right] \tag{5.7}$$

The equilibrium function is recalculated at every iteration using the updated density and velocity fields.

5.1.4. TRT collision operator

The collision step implemented in the `Lattice` class follows the Two-Relaxation-Time formulation.

For each pair of opposite directions, the distribution functions are separated into symmetric and antisymmetric components:

$$f_i^+ = \frac{1}{2}(f_i + f_{\bar{i}}) \tag{5.8}$$

$$f_i^- = \frac{1}{2}(f_i - f_{\bar{i}}) \tag{5.9}$$

The two components are independently relaxed:

$$f_i^{+,*} = f_i^+ - \omega^+ (f_i^+ - f_i^{eq,+}) \tag{5.10}$$

$$f_i^{-,*} = f_i^- - \omega^- (f_i^- - f_i^{eq,-}) \tag{5.11}$$

The post-collision distributions are reconstructed as:

$$f_i^{post} = f_i^{+,*} + f_i^{-,*} \tag{5.12}$$

Only one member of each opposite velocity pair is explicitly processed to avoid redundant operations.

5.1.5. Streaming procedure

The propagation of particle distributions is performed using a pull streaming scheme.

For each lattice direction, the incoming distribution is obtained from the neighbouring lattice node:

$$f_i(\mathbf{x}, t+1) = f_i^{post}(\mathbf{x} - \mathbf{c}_i, t) \tag{5.13}$$

A temporary distribution field is used during this operation to avoid overwriting values that are still required during the update.

5.2. Boundary class implementation

The `Boundary` class manages all the boundary conditions required by the lid-driven cavity problem.

The implementation includes:

- bounce-back conditions for stationary walls;
- moving-wall treatment for the lid;
- explicit corner handling.

5.2.1. Stationary wall boundary conditions

The bottom, left, and right walls are treated as no-slip stationary boundaries using the bounce-back scheme.

The distribution functions reaching a solid boundary are reflected towards the fluid domain:

$$f_i = f_{\bar{i}} \tag{5.14}$$

This condition imposes zero velocity at the wall.

5.2.2. Moving lid boundary condition

The upper wall moves with constant velocity U_{lid}.

The unknown populations after streaming are reconstructed imposing the desired wall momentum. The correction terms associated with the moving wall are:

$$f_5 = f_6 + \frac{\rho U_{lid}}{6} \tag{5.15}$$

$$f_7 = f_8 - \frac{\rho U_{lid}}{6} \tag{5.16}$$

This allows the transfer of momentum from the moving boundary to the fluid.

5.2.3. Corner treatment

The four corners are treated separately because they are simultaneously affected by two different boundary conditions.

Specific assignments are applied in order to avoid conflicting reflections between adjacent walls.

5.3. Cavity class implementation

The `Cavity` class represents the high-level solver controlling the complete simulation.

It initializes the lattice and boundary objects, manages the time iterations, evaluates convergence, and provides access to the final solution.

5.3.1. Simulation workflow

At each iteration the following operations are performed:

1. computation of macroscopic variables;
2. equilibrium distribution evaluation;
3. TRT collision;
4. streaming;
5. application of boundary conditions;
6. update of macroscopic variables.

This sequence is repeated until the steady-state condition is reached.

5.3.2. Convergence monitoring

The convergence of the solution is monitored through the relative variation of the velocity field between two consecutive iterations:

$$\epsilon = \sqrt{\frac{\sum (u - u^{old})^2}{\sum u^2 + \delta}} \tag{5.17}$$

where δ is a small numerical parameter introduced to avoid division by zero.

The simulation is considered converged when:

$$\epsilon < 10^{-8} \tag{5.18}$$

6 | Conclusions and future developments

This work presented the development and validation of a two-dimensional Lattice Boltzmann Method (LBM) solver for the simulation of the lid-driven cavity flow problem. The implemented numerical model is based on a D2Q9 lattice and a Two-Relaxation-Time (TRT) collision operator, which represents a standard approach in LBM applications involving incompressible viscous flows and complex boundary conditions.

The implemented solver includes the main components required for a complete LBM simulation: initialization of the distribution functions, macroscopic variables reconstruction, equilibrium distribution calculation, TRT collision, streaming procedure, and application of wall boundary conditions. The lid-driven cavity configuration was selected as a reference problem due to its relevance in computational fluid dynamics and because of the availability of benchmark solutions reported by Ghia et al.

The comparison with the reference solution has shown that, for low and moderate Reynolds numbers, the proposed LBM formulation is able to reproduce the main characteristics of the flow field and provides velocity profiles in agreement with the benchmark data. This confirms the capability of the D2Q9-TRT model to correctly recover the expected fluid behaviour in the range where the assumptions underlying the LBM formulation are satisfied.

It is important to highlight that the comparison with Ghia et al. does not represent a comparison between two identical numerical approaches. The reference solution was obtained through a conventional Navier–Stokes solver based on a streamfunction–vorticity formulation and multigrid acceleration, whereas the present work employs a mesoscopic LBM approach. Therefore, the benchmark is used as a validation of the resulting velocity field rather than as a direct comparison between numerical algorithms.

The adopted LBM parameters, including the D2Q9 lattice, the TRT collision operator, and the standard relation between viscosity and relaxation parameter, represent commonly used choices in the literature. Nevertheless, when increasing the Reynolds number, the numerical difficulty of the problem increases significantly. In particular, for high Reynolds number cases, the required lattice viscosity becomes very small and the relaxation parameter ω^+ approaches the stability limit of the LBM formulation. Moreover, maintaining the low-Mach-number condition required by weakly compressible LBM approaches becomes more challenging.

For the considered grid resolution and numerical parameters, the case $Re = 10000$ did not reach a stable converged solution. This behaviour should not be interpreted as an intrinsic limitation of the TRT model, which is widely adopted also for high-Reynolds-number simulations, but rather as a consequence of the combined effect of grid resolution, relaxation parameters, and the numerical constraints associated with the LBM approximation. A meaningful comparison with Ghia et al. at very high Reynolds numbers would therefore require a different numerical setup, for example a finer lattice resolution, optimized TRT parameters, or more advanced collision models.

From a computational point of view, the implemented solver has been parallelized using OpenMP. The most computationally demanding operations, namely the computation of macroscopic variables, equilibrium distribution, collision step, and streaming procedure, have been parallelized through shared-memory parallelism. This approach allows an efficient exploitation of multicore processors while maintaining a relatively simple implementation.

A possible future development of the solver would be the extension from OpenMP to a distributed-memory parallelization based on MPI. The LBM method is particularly suitable for this approach because of its local nature: each lattice node requires information only from neighbouring nodes. The computational domain could therefore be divided into sub-domains assigned to different processors, with communication limited to the exchange of distribution functions at the interfaces between neighbouring regions.

The introduction of MPI parallelization would not directly modify the physical formulation of the LBM model, but it would allow the simulation of significantly larger computational domains. This improvement would be particularly beneficial for high-Reynolds-number flows, where finer spatial resolutions are required to maintain numerical stability and accuracy. Furthermore, a distributed implementation would enable the study of a wider range of parameters and the execution of multiple simulations for systematic validation purposes.

Additional future developments could include the implementation of higher-order boundary conditions, the adoption of Multiple-Relaxation- Time (MRT) collision models, and the investigation of adaptive grid strategies. These improvements would further extend the applicability of the solver to more complex fluid-dynamic problems while preserving the advantages of the LBM approach.

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A | Physical overview of the lid-driven cavity problem

The lid-driven cavity flow represents one of the most widely used benchmark problems in computational fluid dynamics. Despite its simple geometrical configuration, it contains the main physical mechanisms governing viscous incompressible flows, including the transport of momentum by advection, viscous diffusion, the interaction with solid boundaries, and the formation of recirculating regions.

The problem consists of a square cavity completely filled with an incompressible fluid. The bottom wall and the two vertical walls remain stationary, while the upper wall moves at a constant tangential velocity U_{lid}. Although the geometry is particularly simple, no analytical solution of the complete Navier–Stokes equations is available, making this configuration an ideal benchmark for validating numerical methods.

The physical mechanism driving the flow originates from the no-slip boundary condition imposed on the moving lid. The fluid particles in contact with the upper wall acquire the same horizontal velocity, whereas the fluid farther from the lid is initially at rest. Viscous stresses progressively transfer momentum from the moving wall toward the interior of the cavity. This momentum diffusion generates a large clockwise recirculating motion, commonly referred to as the *primary vortex*, which occupies most of the computational domain.

As the circulating fluid approaches the stationary walls, the no-slip condition forces the velocity to vanish. Consequently, strong velocity gradients develop close to the boundaries, where viscous stresses become particularly intense. The interaction between the global circulation and these boundary layers is responsible for the appearance of additional localized recirculation regions, usually called *secondary vortices*. Their size and location strongly depend on the Reynolds number and constitute one of the most important indicators of the accuracy of a numerical simulation.

A.1. Physical meaning of the governing parameters

The behaviour of the flow is mainly controlled by the Reynolds number, defined as

$$Re = \frac{U_{lid} L}{\nu}, \tag{A.1}$$

where

- U_{lid} is the velocity of the moving lid;
- L is the characteristic cavity length;
- ν is the kinematic viscosity of the fluid.

The Reynolds number measures the relative importance of inertial and viscous effects:

$$Re = \frac{\text{inertial forces}}{\text{viscous forces}}. \tag{A.2}$$

For small values of Re, viscous diffusion dominates the transport of momentum. The kinetic energy introduced by the moving lid rapidly diffuses throughout the fluid, producing smooth velocity profiles and relatively thick boundary layers.

As the Reynolds number increases, inertial forces become progressively more important. Momentum is transported over longer distances before being dissipated by viscosity, resulting in stronger recirculation, sharper velocity gradients, thinner boundary layers, and increasingly localized flow structures.

The Reynolds number therefore determines the global balance between momentum diffusion and convection and directly influences the topology of the flow field.

A.2. Evolution of the flow structure with Reynolds number

For low Reynolds numbers ($Re \approx 100$), the cavity is characterized by a single primary vortex occupying most of the domain. Owing to the dominance of viscous diffusion, the vortex remains approximately centred inside the cavity and the velocity field varies smoothly. Boundary layers extend over a significant fraction of the cavity height and the flow exhibits an almost symmetric structure.

As the Reynolds number increases to intermediate values ($Re \approx 400$–1000), inertial effects strengthen the circulation generated by the moving lid. The primary vortex gradually migrates towards the upper part of the cavity, where the momentum introduced by the moving wall is largest. Simultaneously, the interaction between the main circulation and the stationary walls produces small secondary vortices near the lower corners. These vortices originate from the local reversal of the flow induced by the confinement of the recirculating motion.

For higher Reynolds numbers ($Re > 3000$), the flow becomes increasingly dominated by inertia. The primary vortex moves further towards the upper-central region of the cavity and becomes more compact. The boundary layers adjacent to the stationary walls become considerably thinner, concentrating large velocity gradients within a very limited region.

At the same time, the secondary vortices located near the lower corners increase both in size and intensity. Additional smaller vortical structures may also develop close to the upper corners, where the moving lid meets the stationary vertical walls. These corner vortices result from the interaction between the high-speed lid motion and the abrupt change imposed by the no-slip condition.

The overall flow therefore evolves from a relatively simple viscous circulation towards a hierarchical vortex system composed of one dominant recirculation and several smaller secondary structures. Their accurate prediction requires an increasingly refined numerical resolution as the Reynolds number grows.

A.3. Relation between physical parameters and LBM limitations

In the Lattice Boltzmann formulation, the kinematic viscosity is related to the relaxation parameter governing the collision process. For the TRT model employed in the present work,

$$\nu = c_s^2 \left(\frac{1}{\omega^+} - \frac{1}{2} \right), \tag{A.3}$$

where ω^+ denotes the symmetric relaxation parameter.

Increasing the Reynolds number while keeping the lattice resolution fixed necessarily implies reducing the physical viscosity. As a consequence, the relaxation parameter approaches its upper stability limit,

$$\omega^+ \to 2.$$

When this condition is approached, viscous dissipation becomes extremely small and the collision operator is no longer able to damp the numerical errors generated by the streaming process and by the boundary treatment. The simulation consequently becomes increasingly sensitive to round-off errors and spatial discretization.

Furthermore, the Lattice Boltzmann Method is derived under the assumption of weak compressibility. The characteristic flow velocity must therefore remain much smaller than the lattice speed of sound,

$$Ma = \frac{U}{c_s} \ll 1, \tag{A.4}$$

where Ma denotes the Mach number.

The simultaneous requirement of maintaining both a sufficiently low Mach number and a relaxation parameter sufficiently far from its stability limit becomes increasingly difficult to satisfy at high Reynolds numbers, unless the computational grid is refined.

From a physical viewpoint, increasing the Reynolds number also produces thinner viscous boundary layers adjacent to the cavity walls. These layers contain the largest velocity gradients of the entire flow and are responsible for the generation of the secondary vortices. If the grid spacing is not sufficiently small compared with the boundary-layer thickness, only a limited number of lattice nodes are available to resolve these gradients. The resulting loss of spatial resolution reduces the accuracy of the momentum exchange between neighbouring nodes, leading to larger numerical errors in the vicinity of the walls.

For this reason, simulations performed at increasingly large Reynolds numbers generally require finer computational grids in order to preserve both numerical stability and physical accuracy.

A.4. Reference benchmark and validation

The lid-driven cavity problem has been extensively investigated in the literature and represents one of the standard validation benchmarks for computational fluid dynamics. In particular, the velocity profiles published by Ghia et al. are commonly employed to assess the accuracy of new numerical methods.

The comparison with these benchmark profiles validates the macroscopic flow solution predicted by the implemented solver. It should be noted, however, that the reference solution was obtained by solving the incompressible Navier–Stokes equations through a multigrid finite difference method, whereas the present work employs a mesoscopic Lattice Boltzmann formulation. Consequently, the comparison validates the physical consistency of the computed flow field rather than the agreement between identical numerical formulations.

For low and moderate Reynolds numbers, the assumptions underlying the Lattice Boltzmann Method remain fully satisfied and the numerical solution reproduces the benchmark velocity profiles with excellent accuracy. At higher Reynolds numbers, thinner boundary layers, increasingly localized velocity gradients and relaxation parameters approaching their stability limits require either finer computational grids or more advanced numerical formulations.

For a more detailed physical analysis of the flow structures, Ghia et al. also report contour maps of the streamfunction and the vorticity. These quantities provide a much more complete description of the recirculation zones and of the evolution of the vortex system than the centreline velocity profiles alone, and therefore represent the natural reference for an in-depth physical interpretation of the cavity flow.

A.5. Comparison between BGK and TRT collision models

The collision operator represents one of the fundamental components of the Lattice Boltzmann Method, since it controls the relaxation of the particle distribution functions towards the local equilibrium state. Different collision models can be adopted depending on the required balance between computational efficiency, accuracy and numerical stability.

The simplest and most widely used formulation is the single relaxation time model, commonly known as the Bhatnagar–Gross–Krook (BGK) model. In this approach, all non-equilibrium components of the distribution function are relaxed towards equilibrium using a single relaxation parameter:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \omega \left[f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t) \right]. \tag{A.5}$$

The relaxation parameter ω is directly related to the kinematic viscosity of the fluid through

$$\nu = c_s^2 \left(\frac{1}{\omega} - \frac{1}{2} \right), \tag{A.6}$$

where c_s is the lattice speed of sound. The BGK formulation is particularly attractive because of its simplicity and low computational cost. However, the use of a single relaxation time imposes the same dissipation rate to all kinetic modes of the distribution function. Consequently, when simulations are performed at high Reynolds numbers, the relaxation parameter approaches the stability limit $\omega = 2$, leading to reduced numerical damping and increased sensitivity to discretization errors.

To overcome this limitation, the Two Relaxation Time (TRT) model introduces two independent relaxation parameters by separating the distribution function into symmetric and antisymmetric components with respect to opposite lattice directions.

For a pair of opposite directions i and $\bar{i}$, the symmetric and antisymmetric contributions are defined as

$$f_i^+ = \frac{1}{2}(f_i + f_{\bar{i}}), \tag{A.7}$$

and

$$f_i^- = \frac{1}{2}(f_i - f_{\bar{i}}). \tag{A.8}$$

The collision step is then performed independently for the two components:

$$f_i^{+,*} = f_i^+ - \omega^+ (f_i^+ - f_i^{eq,+}), \tag{A.9}$$

$$f_i^{-,*} = f_i^- - \omega^- (f_i^- - f_i^{eq,-}). \tag{A.10}$$

The symmetric relaxation parameter ω^+ controls the viscous behaviour of the fluid and is directly related to the kinematic viscosity. Conversely, the antisymmetric relaxation parameter ω^- does not directly influence the macroscopic viscosity and can be adjusted to improve numerical stability.

In the present implementation, the TRT model is adopted because it provides improved stability properties compared with the BGK formulation, especially for confined flows involving solid boundaries. The second relaxation parameter is selected through the so-called "magic parameter" relation:

$$\Lambda = \left(\frac{1}{\omega^+} - \frac{1}{2} \right) \left(\frac{1}{\omega^-} - \frac{1}{2} \right), \tag{A.11}$$

with

$$\Lambda = \frac{3}{16}, \tag{A.12}$$

which is a commonly adopted value for TRT schemes applied to no-slip boundary conditions.

Compared with BGK, the TRT formulation introduces a slightly higher computational cost because each collision requires the treatment of two relaxation processes instead of one. However, this additional cost is limited and provides a significant improvement in stability and boundary accuracy.

For the lid-driven cavity problem considered in this work, the advantage of TRT becomes particularly relevant because the solution contains strong velocity gradients close to the walls. Nevertheless, TRT does not remove the fundamental limitations of the Lattice Boltzmann approach at very high Reynolds numbers. When the viscosity becomes extremely small, the relaxation parameter ω^+ approaches its upper stability limit, and additional measures such as grid refinement or alternative numerical strategies may still be required.

B | Parallelization strategy and computational scalability

The computational cost of Lattice Boltzmann simulations is mainly related to the repeated evolution of the distribution functions over the entire lattice domain. For a two-dimensional D2Q9 model, each lattice node requires the computation of nine particle populations during the collision and streaming procedures. Although the operations performed at each node are relatively simple, the overall computational effort increases significantly when higher spatial resolutions or more complex flow conditions are considered.

For this reason, parallel computing represents an important aspect in the development of efficient LBM solvers. The local nature of the method, where each lattice node interacts only with its neighbouring nodes, makes LBM particularly suitable for parallel implementations.

B.1. OpenMP shared-memory parallelization

The present implementation adopts OpenMP as a parallel programming model. OpenMP allows the execution of independent operations through multiple threads sharing the same memory space.

The most computationally intensive sections of the solver have been parallelized, including:

- reconstruction of macroscopic variables from the distribution functions;
- computation of the equilibrium distribution;
- TRT collision operation;
- streaming procedure.

These operations are performed independently for different lattice nodes, allowing the computational workload to be distributed among multiple threads.

The main advantage of this approach is the reduction of execution time while preserving the original numerical structure of the LBM algorithm. Furthermore, OpenMP represents an effective solution for workstation and desktop architectures equipped with multiple CPU cores.

B.2. Computational resources and scalability

The simulations presented in this work were performed on a personal computer equipped with a quad-core processor. Although the available hardware limits the maximum number of simultaneously active threads, the parallel structure of the solver is not restricted to this specific configuration.

The implemented parallel regions have been designed to exploit larger computational resources by increasing the number of available processing units. On systems equipped with a higher number of cores, the same numerical operations can be distributed among a larger number of threads, reducing the computational cost of large-scale simulations.

The main benefit of parallelization is therefore not only the acceleration of the current simulations, but also the possibility of addressing problems that would otherwise become computationally expensive. Examples include:

- finer spatial grids required for high Reynolds number flows;
- larger computational domains;
- extensive parametric studies;
- more complex physical configurations.

B.3. Possible extension to distributed-memory parallelization

Although OpenMP is effective for shared-memory systems, larger simulations may require distributed-memory parallelization. A natural extension of the present solver would therefore consist in the introduction of the Message Passing Interface (MPI).

The Lattice Boltzmann Method is particularly suitable for MPI implementations because of its local computational structure. Each lattice node requires information only from neighbouring nodes, allowing the computational domain to be divided into independent sub-domains assigned to different processors.

During the simulation, processors would exchange only the distribution functions located at the interfaces between adjacent sub-domains. This communication procedure, commonly referred to as halo exchange, allows the numerical scheme to be efficiently distributed while preserving the same physical formulation.

B.4. Expected benefits of MPI implementation

The introduction of MPI would not modify the physical model or the accuracy of the LBM formulation itself. Its main purpose would be to increase the available computational resources and enable simulations with higher spatial and temporal requirements.

The main expected advantages would include:

- simulation of larger lattices while maintaining reasonable computational times;
- improved accuracy for high Reynolds number flows through finer spatial discretization;
- possibility of performing a larger number of simulations for validation and parameter studies;
- exploitation of high-performance computing architectures.

In particular, the capability of increasing the lattice resolution is essential for challenging flow regimes. High Reynolds number simulations require the accurate resolution of thin boundary layers and strong velocity gradients near solid walls. A distributed-memory parallel implementation would therefore provide the computational capability required to investigate such regimes while preserving the advantages of the LBM approach.